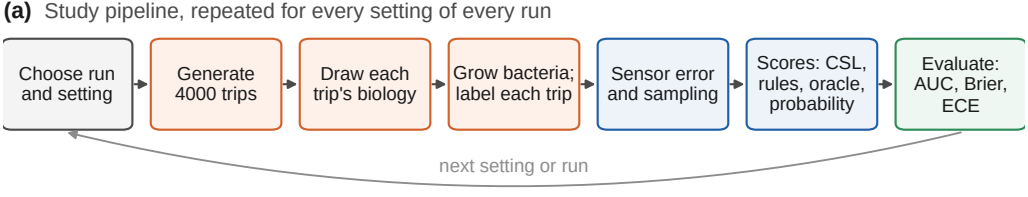
(a) Study pipeline, repeated for every setting of every run
Choose run and setting
Generate 4000 trips
Draw each trip's biology
Grow bacteria; label each trip
Sensor error and sampling
Scores: CSL, rules, oracle, probability
Evaluate: AUC, Brier, ECE
next setting or run
(b) Statistical procedures applied to each setting
Labels y and scores s
AUC by the Mann–Whitney statistic
Bootstrap, 200 resamples: AUC interval
Paired bootstrap, 400 resamples: AUC difference
Brier score and calibration error
Growth-rate fits: Bf, Af, AICc, cross-validation